

# ASSEMBLE AND DISASSEMBLE

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

**ANS:**

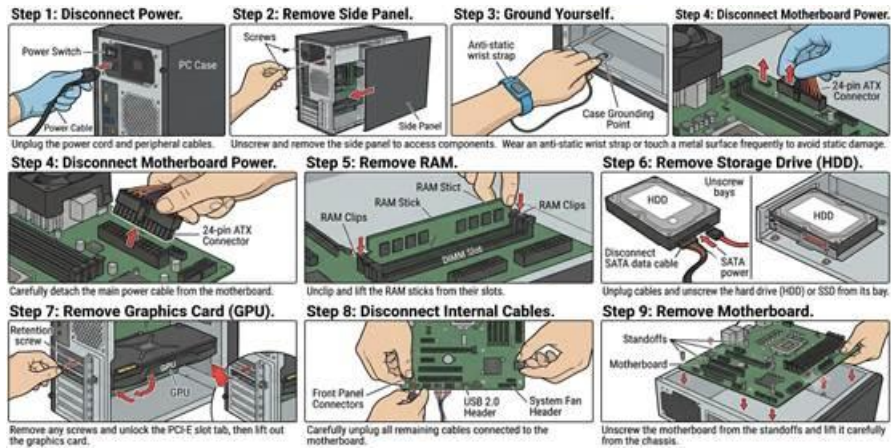
| Component                      | Short description                                                               |
|--------------------------------|---------------------------------------------------------------------------------|
| 1. Motherboard                 | The main circuit board that connects all the components of the computer.        |
| 2. CPU (processor)             | The brain of the computer that performs calculations and executes instructions. |
| 3. CPU cooler (fan/heatsink)   | Keeps the processor cool by removing excess heat.                               |
| 4. RAM (memory)                | Stores temporary data and helps the computer run programs quickly.              |
| 5. Storage (SSD/HDD)           | Stores the operating system, software, and personal files permanently.          |
| 6. Power supply unit           | Converts AC power into DC power and supplies electricity to all components.     |
| 7. Graphics card               | Processes graphics and improves gaming, video editing, and display performance. |
| 8. Case fans                   | Improve airflow inside the cabinet and help keep components cool.               |
| 9. Cables & connectors         | Carry power and transfer data between the computer components.                  |
| 10. Expansion cards (optional) | Add extra features such as Wi-Fi, Bluetooth, or better sound.                   |

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

**Hint:** Include steps like

unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.</em>

**ANS:**



3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

**ANS:**

| Step                                   | Action Performed                                                                  | Why the Order Matters                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1. Install the Motherboard             | Placed the motherboard inside the case and secured it with screws.                | The motherboard must be installed first because all other components connect to it. |
| 2. Install the CPU Cooler              | Attached the CPU cooler and connected its fan cable to the CPU_FAN header.        | The CPU needs proper cooling before the system is powered on.                       |
| 3. Install the RAM                     | Inserted the RAM modules into the correct memory slots until the latches clicked. | RAM should be installed before larger components block access to the slots.         |
| 4. Install the Storage Drive (SSD/HDD) | Mounted the SSD/HDD and connected the SATA data and power cables.                 | Storage must be connected so the operating system and files can be accessed.        |

|                                           |                                                                                                  |                                                                                               |
|-------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>5. Install the Graphics Card (GPU)</b> | Inserted the graphics card into the PCIe slot and secured it with screws.                        | The GPU is installed after the motherboard and RAM to avoid obstructing access.               |
| <b>6. Connect Power Cables</b>            | Connected the 24-pin motherboard cable, CPU power cable, GPU power cable, and SATA power cables. | All components require power connections before the PC can start.                             |
| <b>7. Connect Front Panel Cables</b>      | Connected the Power Switch, Reset Switch, Power LED, HDD LED, USB, and Audio cables.             | These cables enable the case buttons, LEDs, USB ports, and audio ports to function correctly. |
| <b>8. Check All Connections</b>           | Verified that every cable and component was securely connected.                                  | Prevents loose connections and reduces the chance of boot errors.                             |
| <b>9. Close the Side Panel</b>            | Reattached the side panel and tightened the screws.                                              | Protects the internal components and maintains proper airflow.                                |
| <b>10. Power On and Test</b>              | Connected the power cable and turned on the PC to confirm successful reassembly.                 | Final testing ensures that all components are installed and functioning correctly.            |

**4.**Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

**ANS:**

| <b>Safety Precaution / Tool</b>  | <b>Why It Is Important</b>                                                                  |
|----------------------------------|---------------------------------------------------------------------------------------------|
| Turn Off the PC                  | Prevents electrical shock and protects the components from damage.                          |
| Unplug the Power Cable           | Ensures that no electricity is flowing through the computer during assembly or disassembly. |
| Wear an Anti-Static Wrist Strap  | Prevents electrostatic discharge (ESD), which can damage sensitive electronic components.   |
| Work on a Clean, Flat Surface    | Reduces the risk of losing screws and prevents accidental damage to components.             |
| Use a Phillips Screwdriver       | Required to remove and tighten most screws inside the computer cabinet.                     |
| Keep Screws in a Small Container | Prevents screws from getting lost and makes reassembly easier.                              |
| Handle Components by the Edges   | Avoids touching electronic circuits and reduces the risk of static damage.                  |
| Disconnect Cables Carefully      | Prevents damage to connectors, cables, and motherboard ports.                               |
| Use Good Lighting                | Makes it easier to identify components and connect cables correctly.                        |
| Do Not Force Components          | Prevents bending pins, breaking slots, or damaging hardware during installation.            |